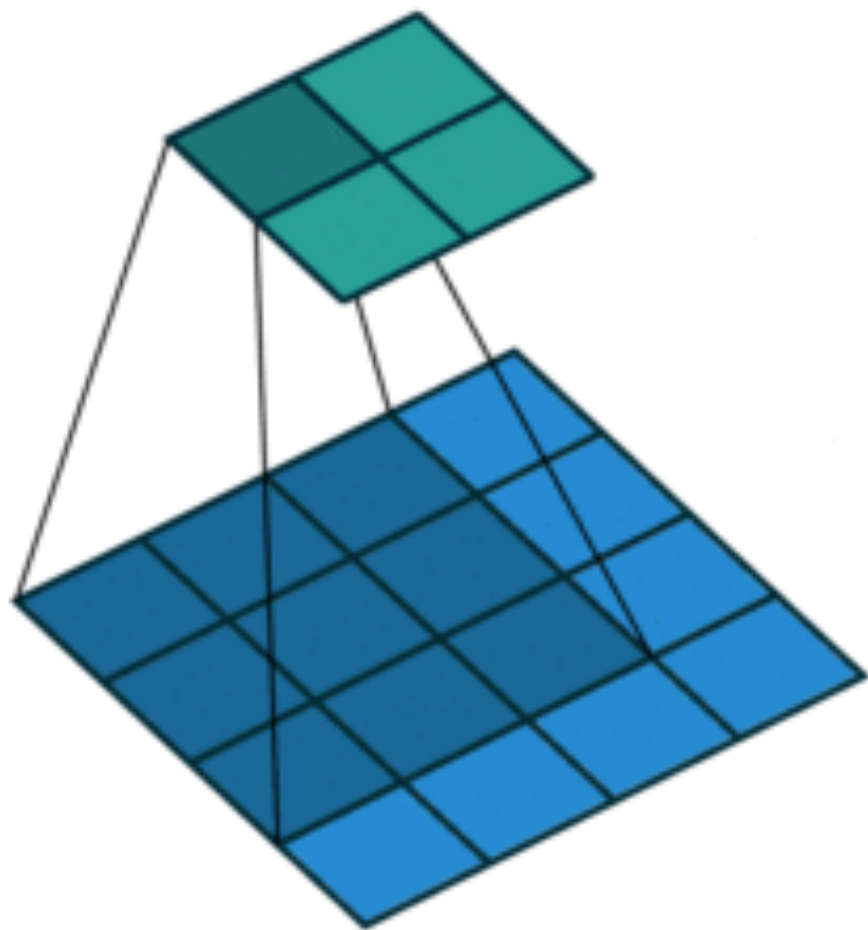


step by step

- Position (1,1), kernel aligned with top-left 3x3 patch of input
- Position (1,2), slide one column right
- Position (2,1), left edge, one row down
- Position (2,2), right edge, one row down

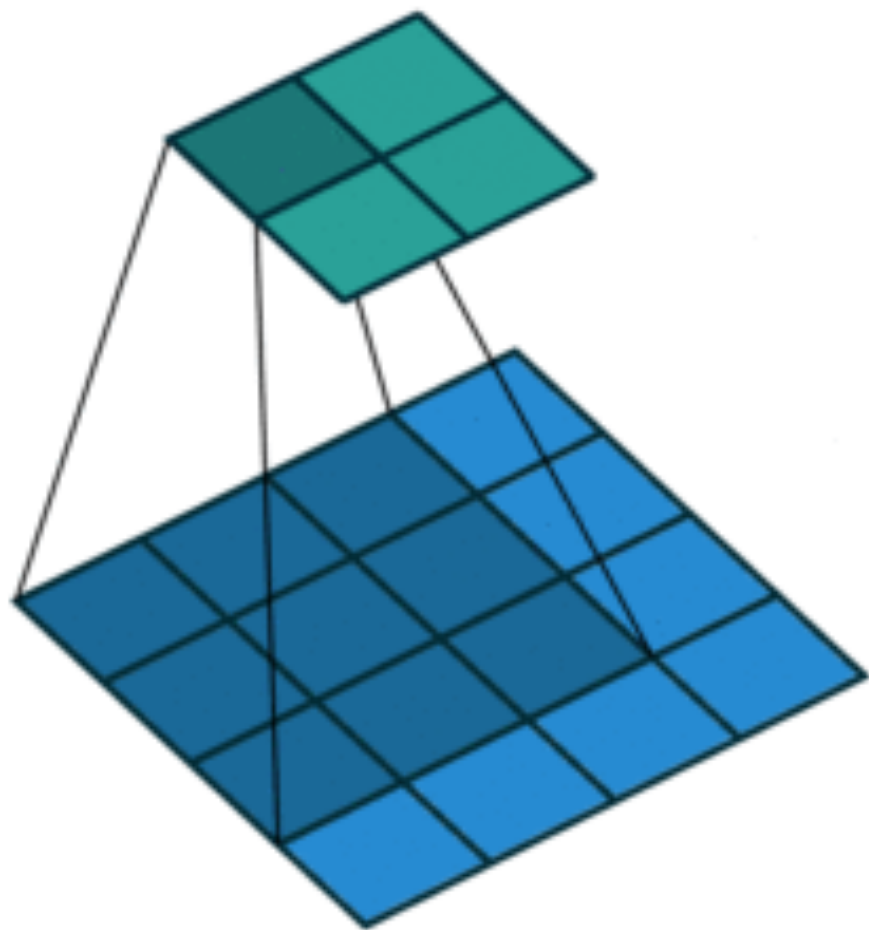


$$Y_{11} = x_{11}k_{11} + x_{12}k_{12} + x_{13}k_{13} + x_{21}k_{21} + x_{22}k_{22} + x_{23}k_{23} + x_{31}k_{31} + x_{32}k_{32} + x_{33}k_{33}$$

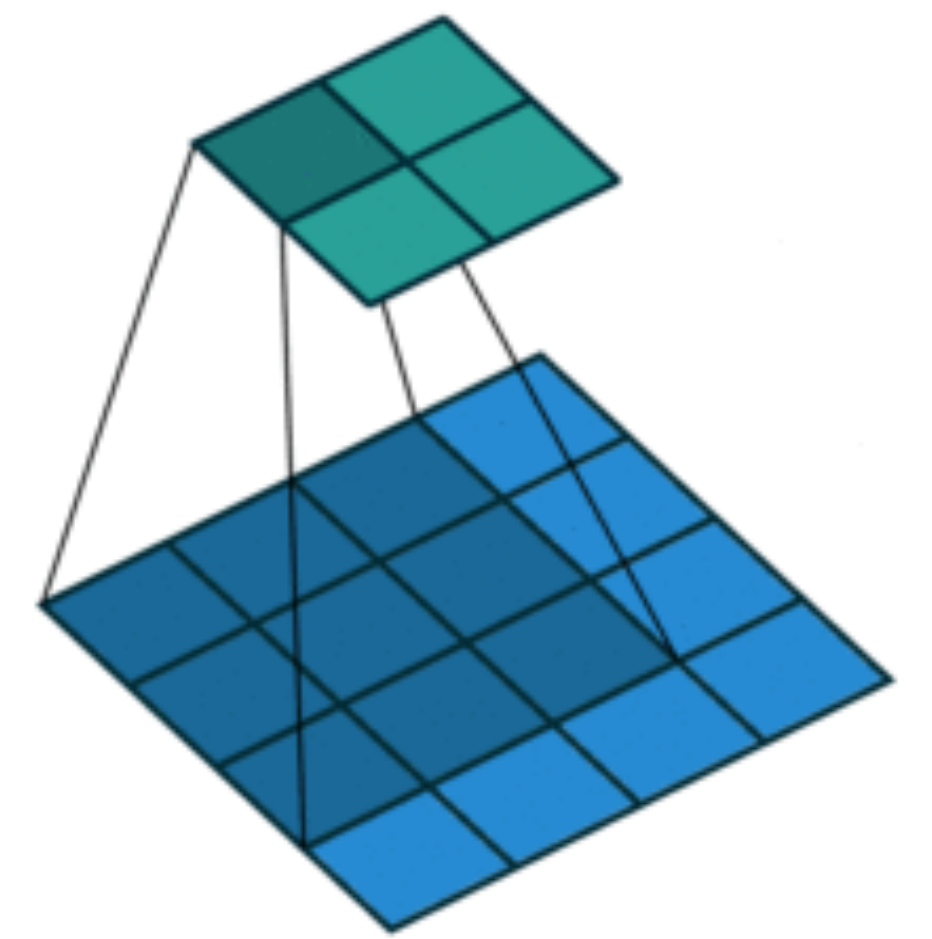
$$Y_{12} = x_{12}k_{11} + x_{13}k_{12} + x_{14}k_{13} + x_{22}k_{21} + x_{23}k_{22} + x_{24}k_{23} + x_{32}k_{31} + x_{33}k_{32} + x_{34}k_{33}$$

$$Y_{21} = x_{21}k_{11} + x_{22}k_{12} + x_{23}k_{13} + x_{31}k_{21} + x_{32}k_{22} + x_{33}k_{23} + x_{41}k_{31} + x_{42}k_{32} + x_{43}k_{33}$$

$$Y_2 = x_{22}k_{11} + x_{23}k_{12} + x_{24}k_{13} + x_{32}k_{21} + x_{33}k_{22} + x_{34}k_{23} + x_{42}k_{31} + x_{43}k_{32} + x_{44}k_{33}$$



Step by step



- Position (1,1), kernel aligned with top-left 3x3 patch of input

$$Y_{11} = x_{11}k_{11} + x_{12}k_{12} + x_{13}k_{13} + x_{21}k_{21} + x_{22}k_{22} + x_{23}k_{23} + x_{31}k_{31} + x_{32}k_{32} + x_{33}k_{33}$$

- Position (1,2), slide one column right

$$Y_{12} = x_{12}k_{11} + x_{13}k_{12} + x_{14}k_{13} + x_{22}k_{21} + x_{23}k_{22} + x_{24}k_{23} + x_{32}k_{31} + x_{33}k_{32} + x_{34}k_{33}$$

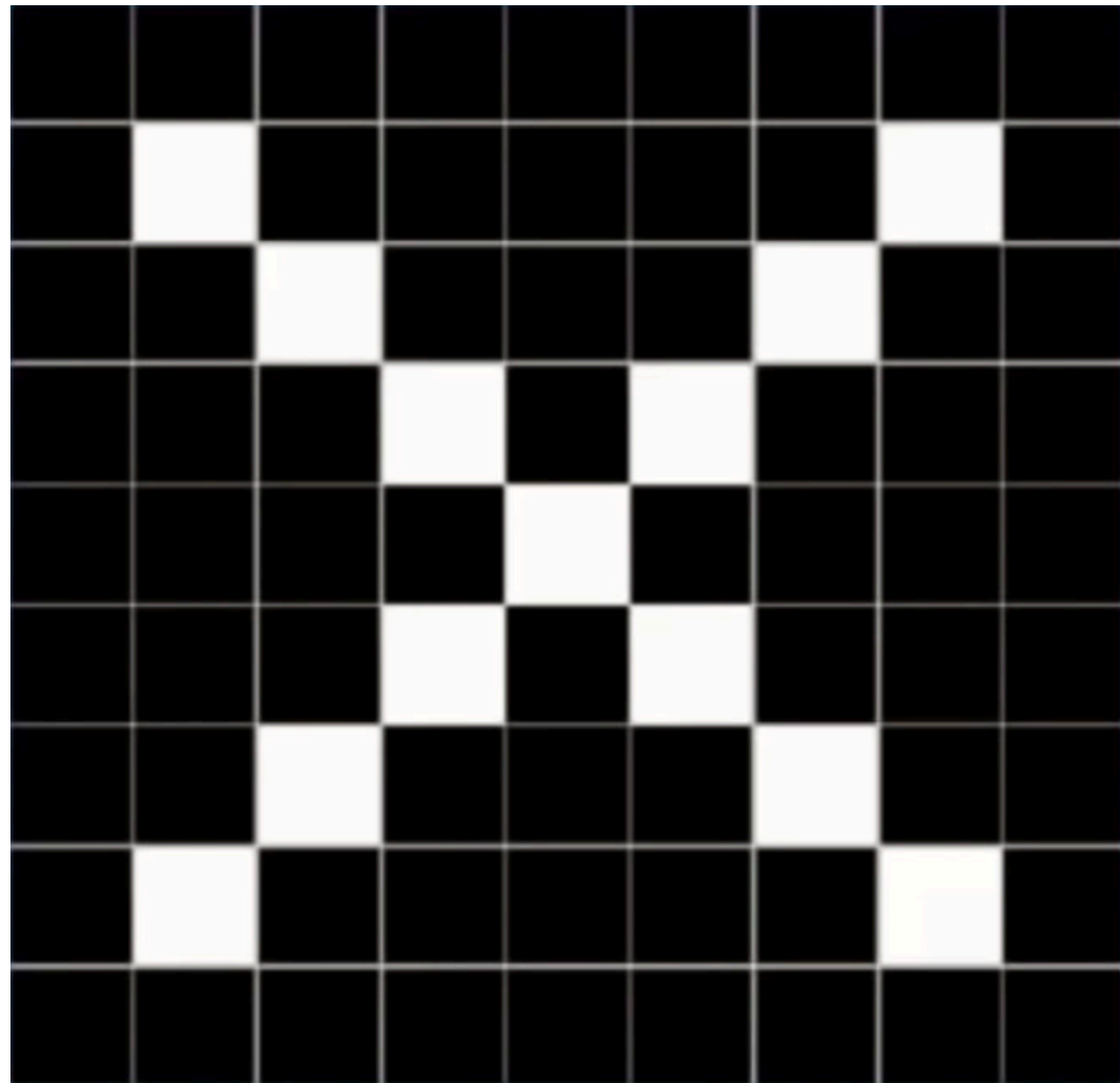
- Position (2,1), left edge, one row down

$$Y_{21} = x_{21}k_{11} + x_{22}k_{12} + x_{23}k_{13} + x_{31}k_{21} + x_{32}k_{22} + x_{33}k_{23} + x_{41}k_{31} + x_{42}k_{32} + x_{43}k_{33}$$

- Position (2,2), right edge, one row down

$$Y_{22} = x_{22}k_{11} + x_{23}k_{12} + x_{24}k_{13} + x_{32}k_{21} + x_{33}k_{22} + x_{34}k_{23} + x_{42}k_{31} + x_{43}k_{32} + x_{44}k_{33}$$

Convolution can be used to find local structure



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